A Crash Course in Forecast Reconciliation

杨杨卓然
Yangzhuoran Fin Yang

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About Me

Yangzhuoran Fin Yang

Assistant professor in Statistical Learning
Department of Data Analytics and Digitalisation
Maastricht University, the Netherlands

- Time series forecasting
- Outlier dectection
- R packages

yangzhuoranyang.com
yangzhuoran.yang@maastrichtuniversity.nl

Outline

1 Hierarchical and grouped time series

2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

5 Parallel and extensions

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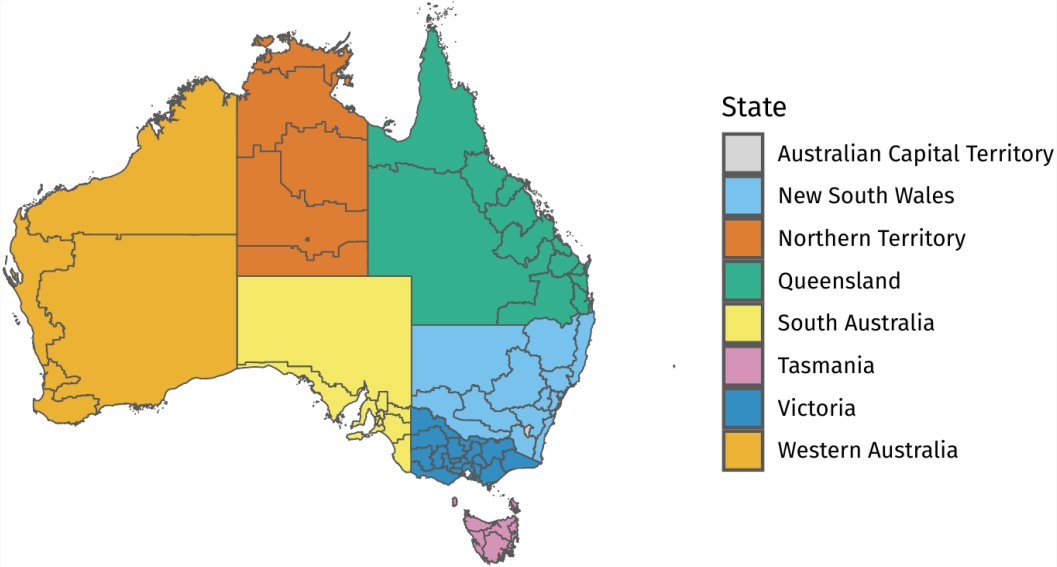
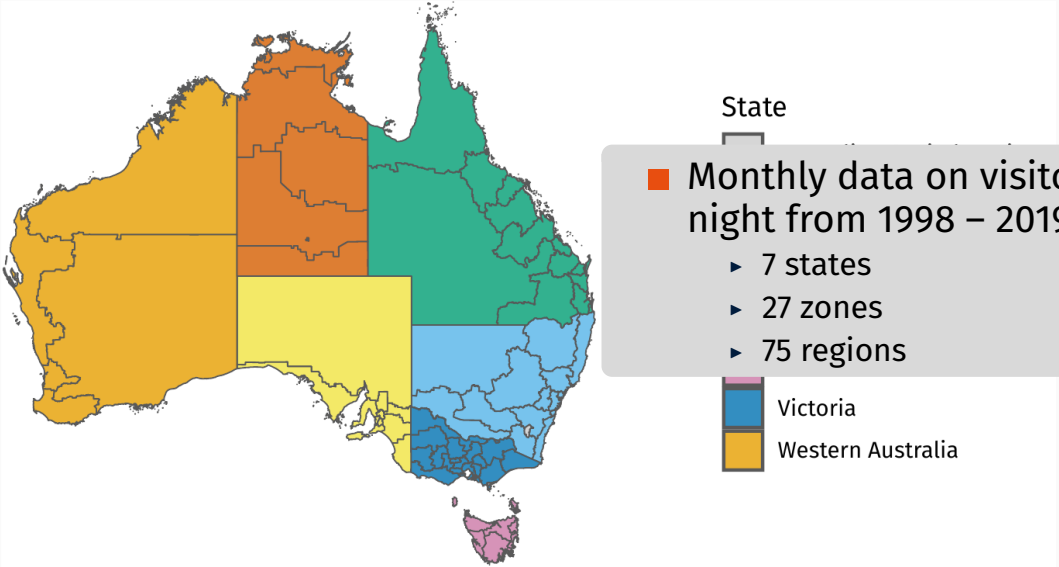
Australian tourism regions

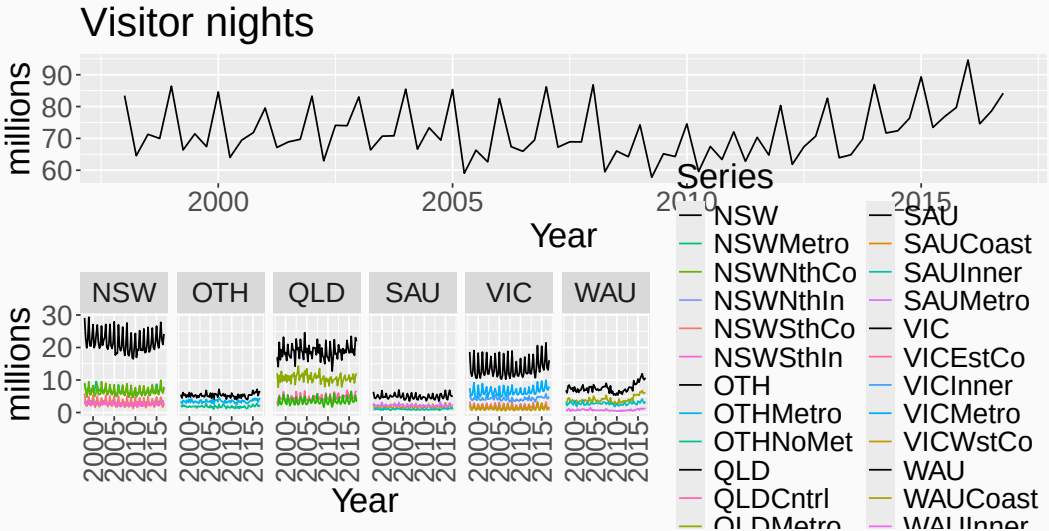


Australian tourism regions



- Monthly data on visitor night from 1998 – 2019
  - 7 states
  - 27 zones
  - 75 regions

Australian tourism data

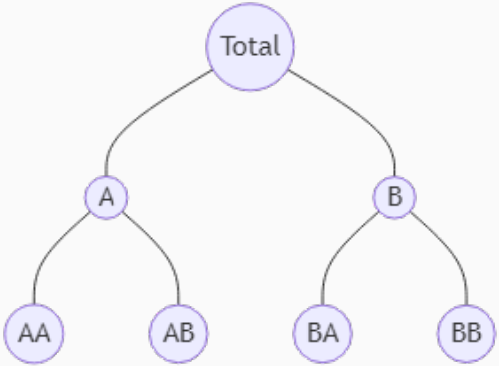


Australian tourism data

```
tsibble::tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:       Region, State, Purpose [304]
   Quarter Region   State           Purpose  Trips
     <qtr> <chr>    <chr>           <chr>    <dbl>
 1 1998 Q1 Adelaide South Australia Business  135.
 2 1998 Q2 Adelaide South Australia Business  110.
 3 1998 Q3 Adelaide South Australia Business  166.
 4 1998 Q4 Adelaide South Australia Business  127.
 5 1999 Q1 Adelaide South Australia Business  137.
 6 1999 Q2 Adelaide South Australia Business  200.
 7 1999 Q3 Adelaide South Australia Business  169.
 8 1999 Q4 Adelaide South Australia Business  134.
 9 2000 Q1 Adelaide South Australia Business  154.
10 2000 Q2 Adelaide South Australia Business  169.
# i 24,310 more rows
```

Hierarchical time series



$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

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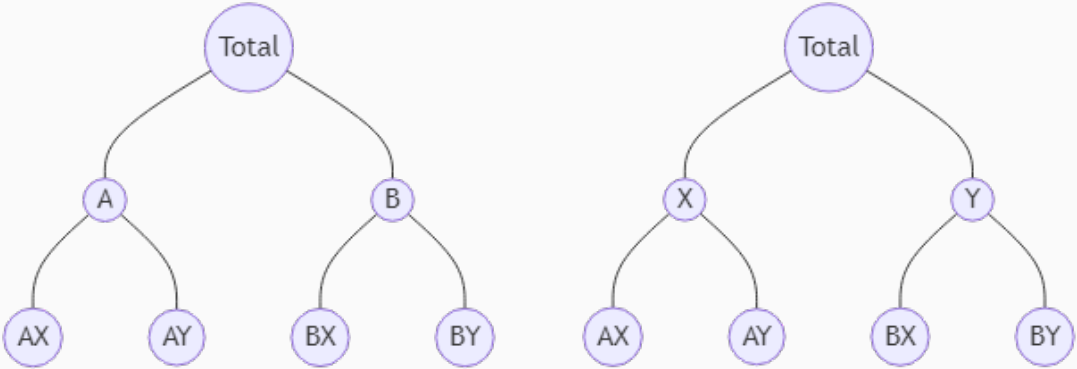
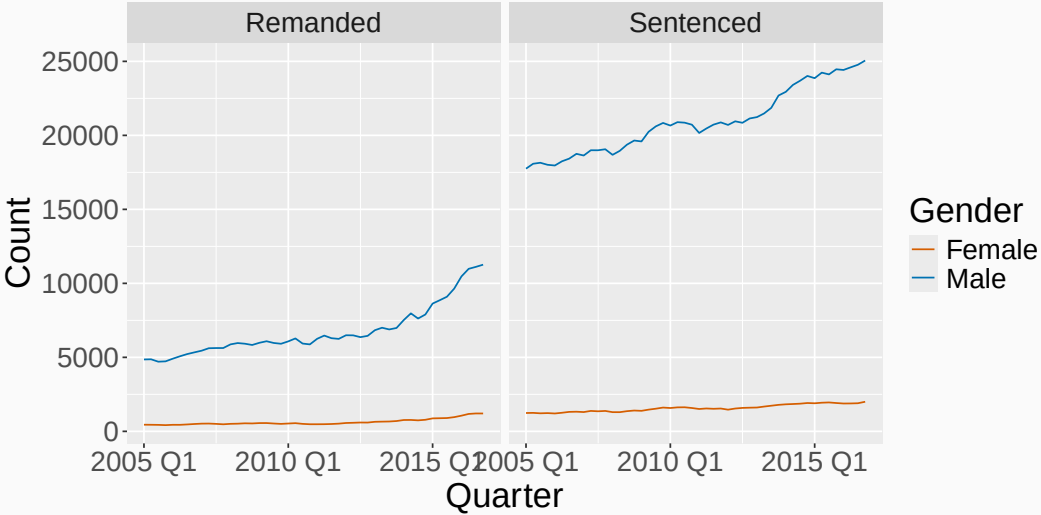
Grouped time series



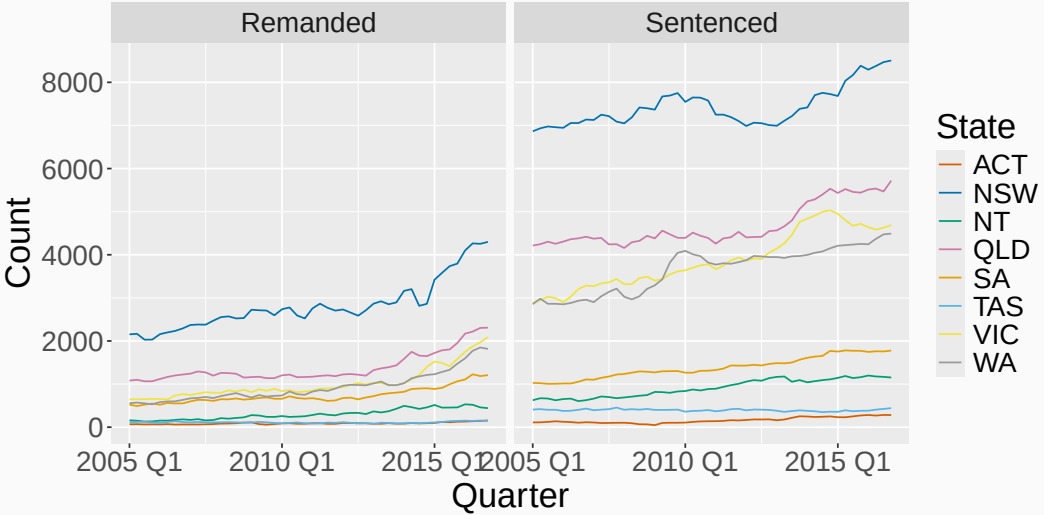
Australian prison population

```
# A tsibble: 3,072 x 6 [1Q]
# Key:       State, Gender, Legal, Indigenous [64]
   Quarter State Gender Legal    Indigenous Count
     <qtr> <chr> <chr>  <chr>    <chr>      <dbl>
 1 2005 Q1 ACT   Female Remanded ATSI           0
 2 2005 Q2 ACT   Female Remanded ATSI           1
 3 2005 Q3 ACT   Female Remanded ATSI           0
 4 2005 Q4 ACT   Female Remanded ATSI           0
 5 2006 Q1 ACT   Female Remanded ATSI           1
 6 2006 Q2 ACT   Female Remanded ATSI           1
 7 2006 Q3 ACT   Female Remanded ATSI           1
 8 2006 Q4 ACT   Female Remanded ATSI           0
 9 2007 Q1 ACT   Female Remanded ATSI           0
10 2007 Q2 ACT   Female Remanded ATSI           1
# i 3,062 more rows
```

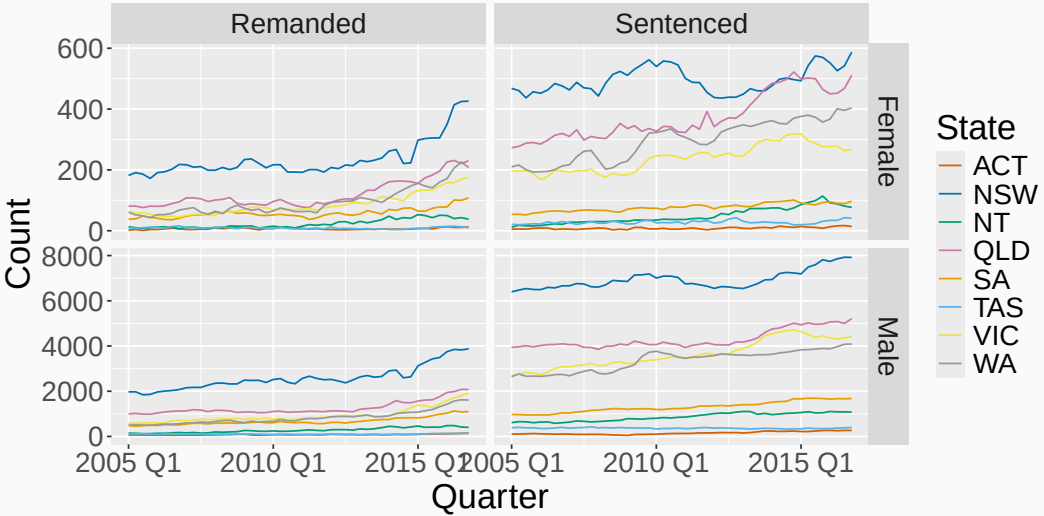
Australian prison population



Australian prison population



Australian prison population



Mixed hierarchical and grouped structure

```
tsibble::tourism
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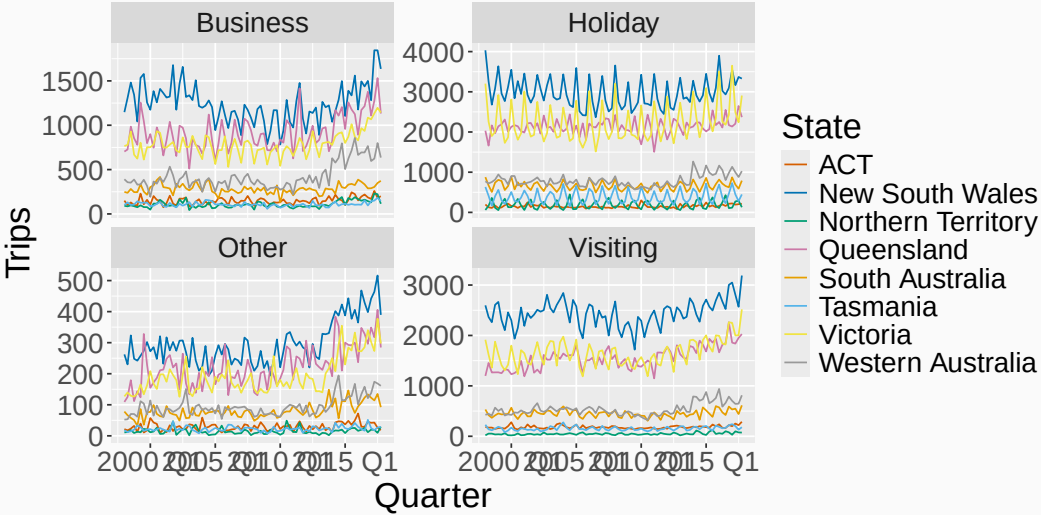
Travel purpose



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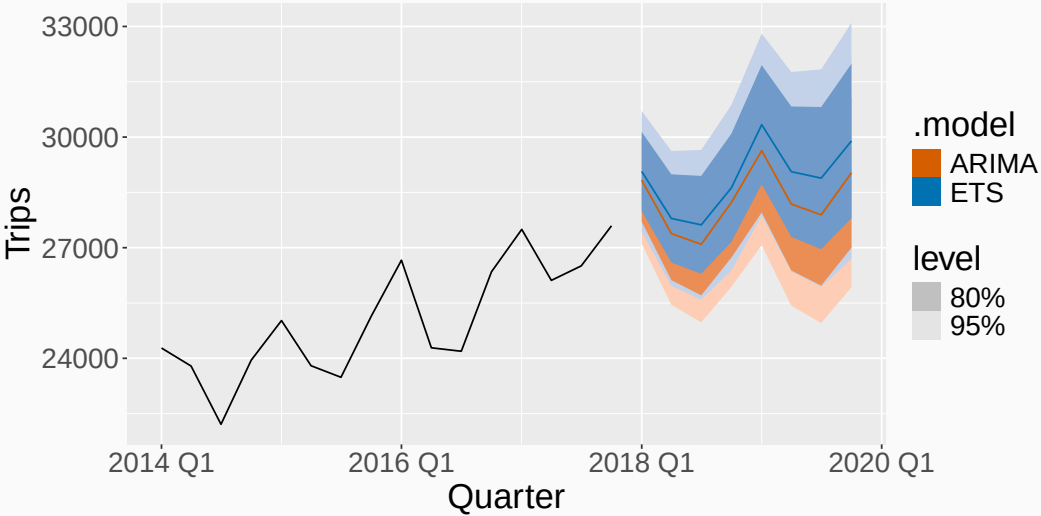
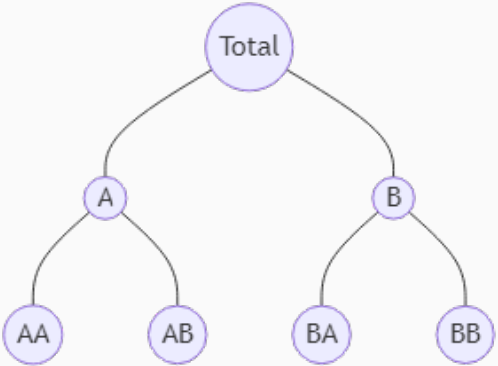
Forecasting method recap



Single level approaches

Goal

Obtain forecasts of every series at all levels, but they need to add up based on the hierarchical structure (**"coherent"**).



- Bottom-up

Single level approaches

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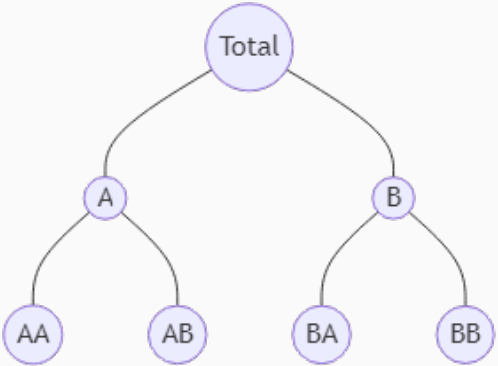

- Bottom-up
- Top-down

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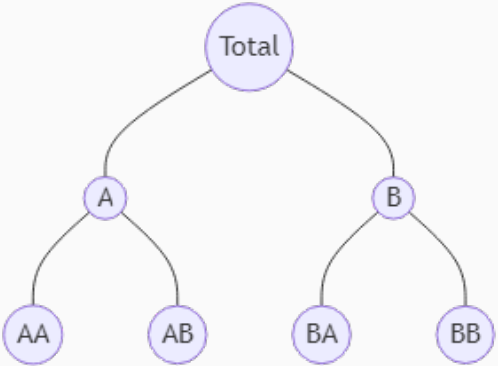
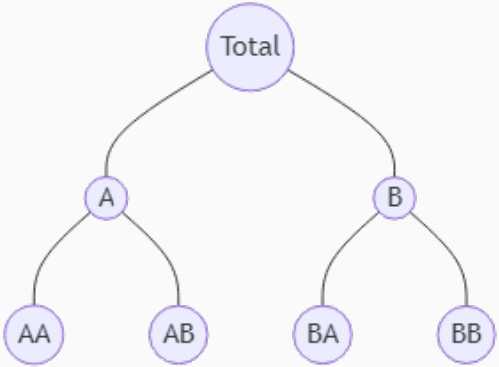


- Bottom-up
- Top-down
- Middle-out

Bottom-up

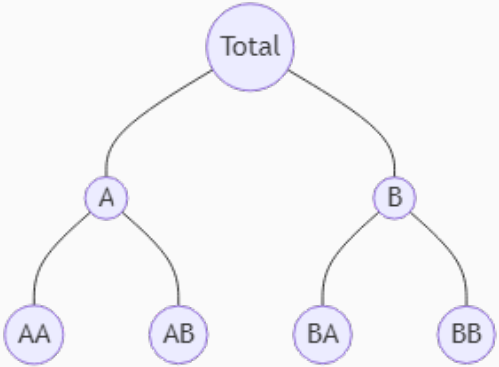


$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

Bottom-up



$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

$$\tilde{y}_t = \hat{y}_{AA,h} + \hat{y}_{AB,h} + \hat{y}_{BA,h} + \hat{y}_{BB,h}$$

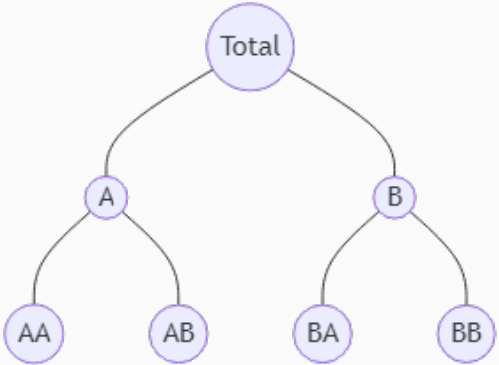
$$\tilde{y}_{A,h} = \hat{y}_{AA,h} + \hat{y}_{AB,h}$$

$$\tilde{y}_{B,h} = \hat{y}_{BA,h} + \hat{y}_{BB,h}$$

Terminology

- $y_{AA,t}$ and friends: the **bottom** level series.
- y_t the **top** level series.
- $y_{AA,t}$: The historical value of series AA at time t.
- $\hat{y}_{AA,h}$: The h-step-ahead forecast for series AA from a forecast model, potentially independently from other series. The **base** forecast.
- $\tilde{y}_{A,h}$: The **reconciled** h-step-ahead forecast for series A that is **coherent**.

Top-down

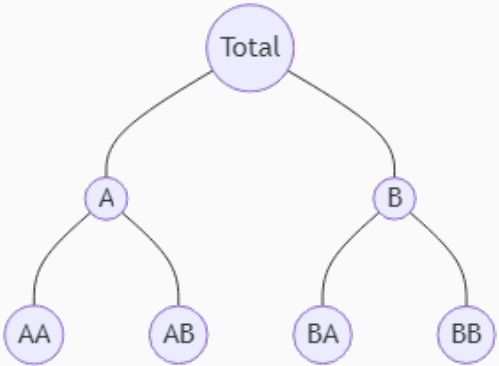


$$y_t = y_{AA,t} + y_{AB,t} + y_{BA,t} + y_{BB,t}$$

$$y_{A,t} = y_{AA,t} + y_{AB,t}$$

$$y_{B,t} = y_{BA,t} + y_{BB,t}$$

Top-down



$$\tilde{y}_{AA,h} = p_{AA}\tilde{y}_t \quad \tilde{y}_{AB,h} = p_{AB}\tilde{y}_t$$

$$\tilde{y}_{BA,h} = p_{BA}\tilde{y}_t \quad \tilde{y}_{BB,h} = p_{BB}\tilde{y}_t$$

$$\tilde{y}_{A,h} = \tilde{y}_{AA,h} + \tilde{y}_{AB,h}$$

$$\tilde{y}_{B,h} = \tilde{y}_{BA,h} + \tilde{y}_{BB,h}$$

What are the proportions?

Average historical proportions

$$p_j = \frac{1}{T}\sum_{t=1}^{T}\frac{y_{j,t}}{y_t}$$

Proportions of the historical averages

$$p_j = \sum_{t=1}^{T}\frac{y_{j,t}}{T}\Big/\sum_{t=1}^{T}\frac{y_t}{T}$$

Forecast proportions

Use the proportion of the base forecasts at a lower level to the aggregate of the base forecasts at this level.

Middle-out

1. Choose a middle level to generate base forecasts
2. Bottom-up upwards
3. Top-down downwards

Middle-out

1. Choose a middle level to generate base forecasts
2. Bottom-up upwards
3. Top-down downwards

Problem

Seems like a waste - only one level is chosen (top, bottom or middle), and the information in other series are not used.

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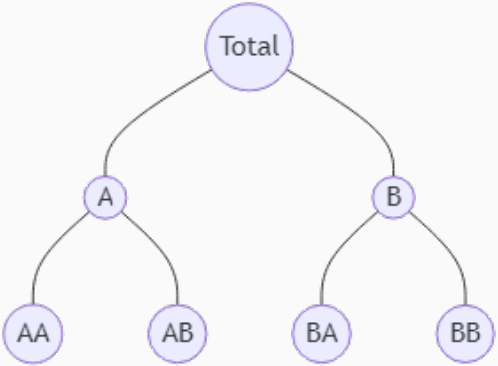
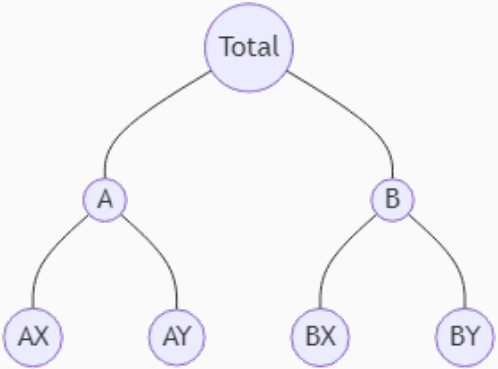
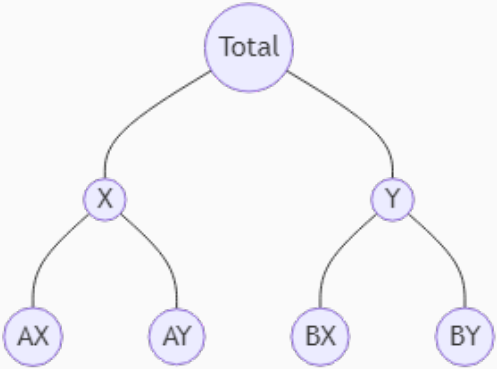
Matrix notation



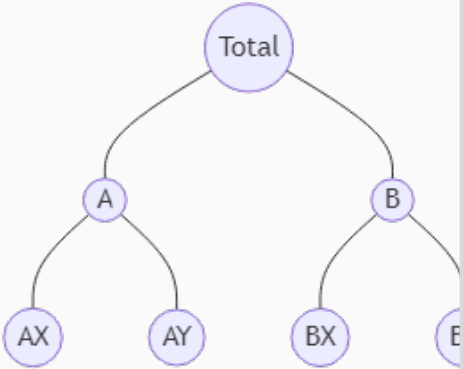
$$\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix}$$

Matrix notation



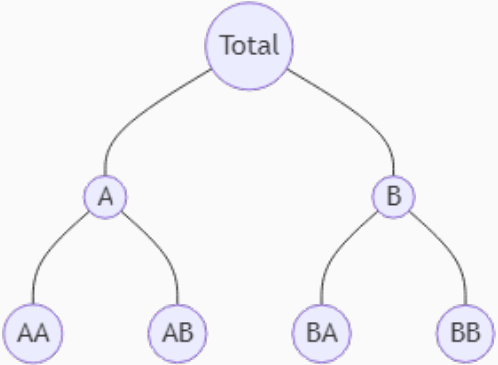


Matrix notation



$$
\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{X,t} \\ y_{Y,t} \\ y_{AX,t} \\ y_{AY,t} \\ y_{BX,t} \\ y_{BY,t} \end{bmatrix}
=
\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} y_{AX,t} \\ y_{AY,t} \\ y_{BX,t} \\ y_{BY,t} \end{bmatrix}
$$

Matrix notation



$$\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix}$$

$$\boldsymbol{y}_t = \boldsymbol{S}\boldsymbol{b}_t = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{b}_t$$

h-step-ahead forecast at time t

$$\boldsymbol{y}_t = \begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} \qquad \hat{\boldsymbol{y}}_{t+h} = \begin{bmatrix} \hat{y}_{t+h} \\ \hat{y}_{A,t+h} \\ \hat{y}_{B,t+h} \\ \hat{y}_{AA,t+h} \\ \hat{y}_{AB,t+h} \\ \hat{y}_{BA,t+h} \\ \hat{y}_{BB,t+h} \end{bmatrix} \qquad \hat{\boldsymbol{b}}_{t+h} = \begin{bmatrix} \hat{y}_{AA,t+h} \\ \hat{y}_{AB,t+h} \\ \hat{y}_{BA,t+h} \\ \hat{y}_{BB,t+h} \end{bmatrix} \qquad \tilde{\boldsymbol{y}}_{t+h} = \begin{bmatrix} \tilde{y}_{t+h} \\ \tilde{y}_{A,t+h} \\ \tilde{y}_{B,t+h} \\ \tilde{y}_{AA,t+h} \\ \tilde{y}_{AB,t+h} \\ \tilde{y}_{BA,t+h} \\ \tilde{y}_{BB,t+h} \end{bmatrix}$$

Forecast reconciliation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{M}\hat{\boldsymbol{y}}_{t+h}$$

Bottom-up

$$\boldsymbol{G} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Top-down

$$\boldsymbol{G} = \begin{bmatrix} p_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_4 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Forecast reconciliation

Limitation

The traditional methods uses only a single level with limited information.

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Goal

Finding the *"optimal"* $\boldsymbol{G}$ matrix to give the most accurate coherent forecasts in.

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Optimality

Optimal in the sense of smallest **forecast error variance** while maintaining **unbiasedness**.

Unbiasedness

If the base forecasts are **unbiased** in the sense that

$$\mathrm{E}(\hat{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t),$$

where $\mathcal{I}_t = \boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_t$ denote data observed up to time t,

we want the reconciled forecast to maintain this property such that

$$\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t).$$

Unbiasedness

Goal

$$\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t).$$

$$\begin{aligned}\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) &= \mathrm{E}(\boldsymbol{SGy}_{t+h}|\mathcal{I}_t) = \boldsymbol{SG}\,\mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t)\\ &= \boldsymbol{SG}\,\mathrm{E}(\boldsymbol{Sb}_{t+h}|\mathcal{I}_t) = \boldsymbol{SGS}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t)\end{aligned}$$

Unbiasedness

Goal

$$\mathrm{E}(\tilde{\boldsymbol{y}}_{t+h}|\mathcal{I}_t) = \mathrm{E}(\boldsymbol{y}_{t+h}|\mathcal{I}_t) = \boldsymbol{S}\,\mathrm{E}(\boldsymbol{b}_{t+h}|\mathcal{I}_t).$$

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Conclusion

When the base forecasts are unbiased, the reconciled forecasts will be unbiased if and only if $\boldsymbol{SGS} = \boldsymbol{S}$, or equivalently, $\boldsymbol{GS} = \boldsymbol{I}$.

Forecast error variance

Denote the base forecast error variance $\mathrm{Var}(\hat{\varepsilon}_{t+h}|\mathcal{I}_t) = \boldsymbol{W}_h$:

$$\boldsymbol{W}_h = \begin{pmatrix} \mathrm{Var}(y_{t+h}) & \mathrm{Cov}(y_{t+h}, y_{A,t+h}) & \cdots & \mathrm{Cov}(y_{t+h}, y_{BB,t+h}) \\ \mathrm{Cov}(y_{A,t+h}, y_{t+h}) & \mathrm{Var}(y_{A,t+h}) & \cdots & \mathrm{Cov}(y_{A,t+h}, y_{BB,t+h}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(y_{BB,t+h}, y_{t+h}) & \mathrm{Cov}(y_{BB,t+h}, y_{A,t+h}) & \cdots & \mathrm{Var}(y_{BB,t+h}) \end{pmatrix}$$

Forecast error variance

Define

$$\hat{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h} \text{ and } \tilde{\varepsilon}_{t+h} = \boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h}$$

Goal

We want to find the forecast error variance $\mathrm{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t)$.

Forecast error variance

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Goal

We want to find the forecast error variance $\text{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t)$.

$$\begin{aligned}
\tilde{\varepsilon}_{t+h} - \hat{\varepsilon}_{t+h} &= \boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h} - \boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h} \\
\tilde{\varepsilon}_{t+h} &= \hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\hat{\boldsymbol{y}}_{t+h} \\
\tilde{\varepsilon}_{t+h} &= \hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})(\boldsymbol{y}_{t+h} - \hat{\varepsilon}_{t+h}) \\
\tilde{\varepsilon}_{t+h} &= \boldsymbol{SG}\hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\boldsymbol{y}_{t+h}
\end{aligned}$$

$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\boldsymbol{S}\boldsymbol{b}_{t+h}$$

$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h},$$

because $\boldsymbol{SGS} = \boldsymbol{S}$ to maintain unbiasedness.

$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h} + (\boldsymbol{I} - \boldsymbol{SG})\boldsymbol{S}\boldsymbol{b}_{t+h}$$
$$\tilde{\varepsilon}_{t+h} = \boldsymbol{SG}\hat{\varepsilon}_{t+h},$$

because $\boldsymbol{SGS} = \boldsymbol{S}$ to maintain unbiasedness.

Forecast error variance

The reconciled forecast error variance is

$$\text{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t) = \text{Var}(\boldsymbol{SG}\hat{\varepsilon}_{t+h}|\mathcal{I}_t) = \boldsymbol{SG}\,\text{Var}(\hat{\varepsilon}_{t+h}|\mathcal{I}_t)\boldsymbol{G}'\boldsymbol{S}' = \boldsymbol{SGW}_h\boldsymbol{G}'\boldsymbol{S}'.$$

Minimising sum of forecast error variances

Trace of a matrix

The trace of a square matrix is the sum of the diagnal elements of that matrix.

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The MinT optimal reconciliation approach

Wickramasuriya et al. (2019) propose the **MinT** method to minimise the trace of the forecast error varianve $\mathrm{Var}(\tilde{\varepsilon}_{t+h}|\mathcal{I}_t)$:

$$\begin{aligned}\boldsymbol{G} &= \arg\min \mathrm{tr}(\boldsymbol{S}\boldsymbol{G}\boldsymbol{W}_h\boldsymbol{G}'\boldsymbol{S}') \text{ s.t. } \boldsymbol{S}\boldsymbol{G}\boldsymbol{S} = \boldsymbol{S} \\ &= (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}\end{aligned}$$

A proof can be found with Ando and Narita (2024).

Minimising ~~sum of~~ forecast error variances

In fact, the MinT solution not only reaches the minimum of the *trace* - the *sum* of the forecast error variances, but also reaches the minimum forecast error varianve individually for each series (Yang et al in a secret working paper).

Alternative representation

Structural representation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{M}\hat{\boldsymbol{y}}_{t+h}$$

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

Mapping matrix:

$$\boldsymbol{M} = \boldsymbol{S}(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

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Zero constrained representation

$$\boldsymbol{y}_t = \boldsymbol{S}\boldsymbol{b}_t = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{I} \end{bmatrix} \boldsymbol{b}_t$$

Constraint matrix:

$$\boldsymbol{C} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \end{bmatrix}$$

$$\boldsymbol{C}\boldsymbol{y}_t = \boldsymbol{0}$$

$$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{W}_h\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{W}\boldsymbol{C}')^{-1}\boldsymbol{C}.$$

Zero constrained representation

THe structural representation

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

requires solving inverse of $\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S}$ with the dimension being **the number of series at the bottom level**.

Zero constrained representation

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requires solving inverse of $\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S}$ with the dimension being **the number of series at the bottom level**. The zero constrained representation (see Di Fonzo and Girolimetto 2023) is **mostly more efficient**.

$$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{W}_h\boldsymbol{C}'(\boldsymbol{C}\boldsymbol{W}\boldsymbol{C}')^{-1}\boldsymbol{C}$$

requires solving inverse of $\boldsymbol{C}\boldsymbol{W}\boldsymbol{C}'$ with the dimension being **the number of series of the upper levels**.

Forecast error variance covariance estimation

To perform forecast reconciliation

$$\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_{t+h},$$

we need

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1},$$

which in turn requires $\boldsymbol{W}_h$.

Problem

How do we get the base forecast error variance $\boldsymbol{W}_h$?

1. OLS

Set $\boldsymbol{W}_h = k_h \boldsymbol{I}$ for all h, where $k_h > 0$. This gives the OLS estimator.

The value of k_h does not matter because it cancells out in $(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$.

1. OLS

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The value of k_h does not matter because it cancells out in $(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$.

- Fast, independent from the data
- Distance reducing
- Good empirical performance
- Effectively says the base forecast error variances are all k_hs and covariances are all 0s.

2. WLS (variance scaling)

Set $\boldsymbol{W}_h = k_h \mathrm{diag}(\hat{\boldsymbol{W}}_1)$ for all h, where $k_h > 0$,

$$\hat{\boldsymbol{W}}_1 = \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_t'.$$

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$$\hat{\boldsymbol{W}}_1 = \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_t'.$$

- Scales the base forecasts using the variance of the residuals
- The WLS estimator using *variance scaling*.
- Assumes base forecast errors are uncorrelated between each other.

3. Structural scaling

Set $\boldsymbol{W}_h = k_h \Lambda$ for all h, where $k_h > 0$, $\Lambda = \mathrm{diag}(\boldsymbol{S}\mathbf{1})$, and $\mathbf{1}$ is a unit vector.

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- Only depends on the hierarchical structure.
- Useful when residuals are not available.

3. Structural scaling

Set $\boldsymbol{W}_h = k_h \Lambda$ for all h, where $k_h > 0$, $\Lambda = \text{diag}(\boldsymbol{S}\mathbf{1})$, and $\mathbf{1}$ is a unit vector.

- Only depends on the hierarchical structure.
- Useful when residuals are not available.
- Assumes bottom-level base forecast errors each have variance k_h and are uncorrelated between each other.
- Each element of Λ contains the number of forecast error variances contributing to each node.

4a. Sample covariance

Set $\boldsymbol{W}_h = k_h \hat{\boldsymbol{W}}_1$ for all h, where $k_h > 0$.

- Direct estimation.
- Assumes only that the error covariance are proportional to each other.

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- Difficult when the structure is large but series are short.

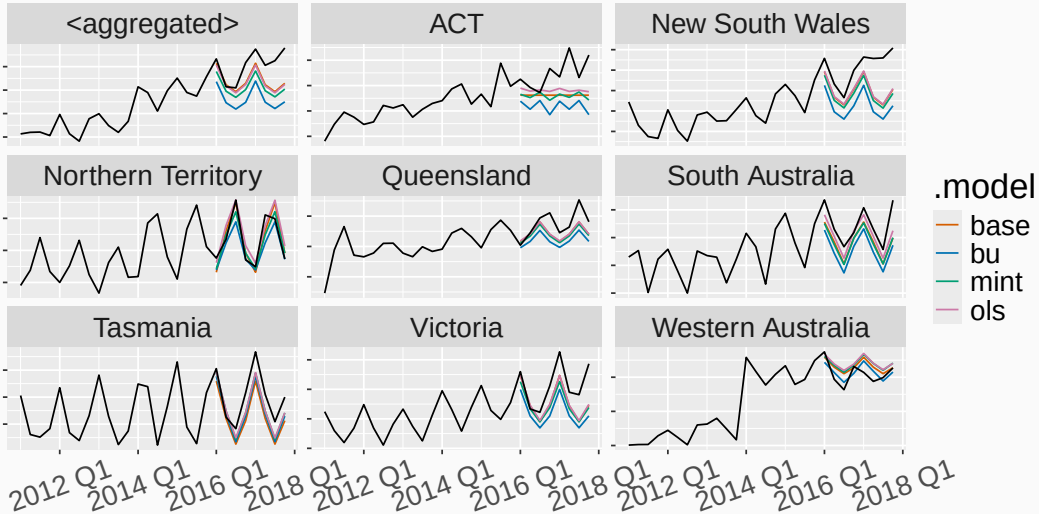
4b. Shrinkage estimator

Shrinking covariance towards zero (Schäfer and Strimmer 2005) and/or **shrinking variance** towards their median (Opgen-Rhein and Strimmer 2007).

- Positive definite, and
- Numerically stable.

`fable` in R supports shrinking covariance.

Forecast Australian domestic tourism



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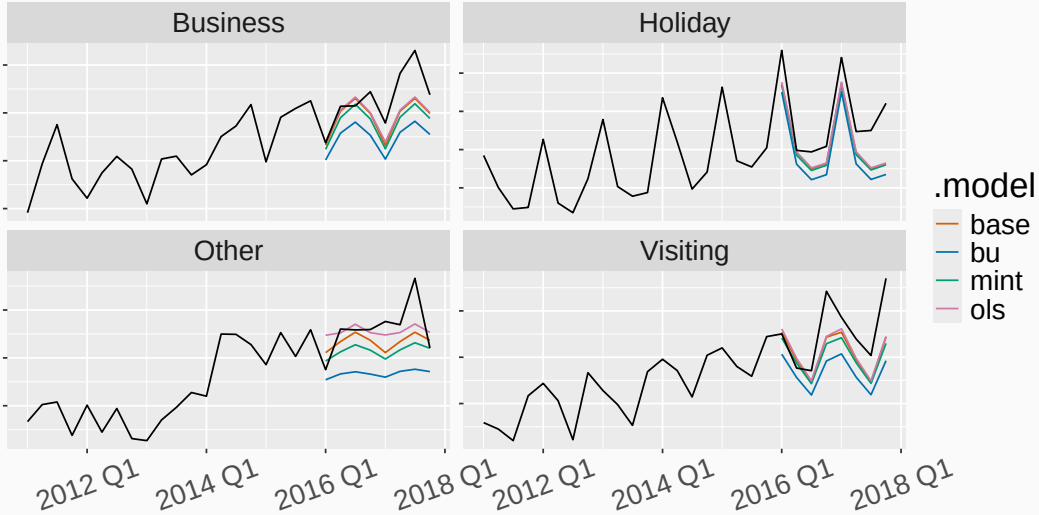

Outline

1 Hierarchical and grouped time series

2 Traditional approaches

3 Forecast reconciliation

4 Projection and geometry

5 Parallel and extensions

Reference

The main reference of this section is Panagiotelis et al. (2021).

OLS recap

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{u}_i$$

$$\boldsymbol{y} = \hat{\beta}_0 + \hat{\beta}_1 \boldsymbol{x} + \hat{\boldsymbol{u}}$$

$$\boldsymbol{y} = \boldsymbol{X}\hat{\beta} + \hat{\boldsymbol{u}}$$

$$\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

OLS recap

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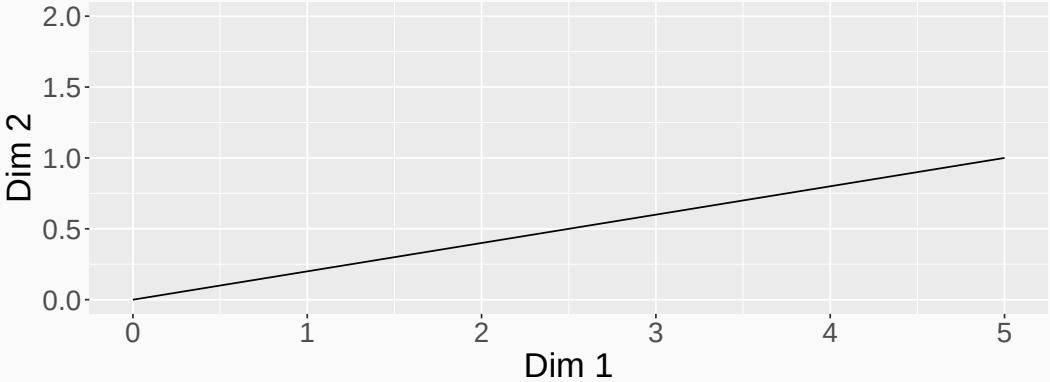
$$\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

$$\begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \hat{\beta}_0 + \begin{bmatrix} 5 \\ 1 \end{bmatrix} \hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix} 1 & 5 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} + \hat{\boldsymbol{u}}$$

OLS geometry - 2 observations

$$\begin{bmatrix}5\\2\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}\hat{\beta}_0 + \begin{bmatrix}5\\1\end{bmatrix}\hat{\beta}_1 + \hat{\boldsymbol{u}} = \begin{bmatrix}1 & 5\\1 & 1\end{bmatrix}\begin{bmatrix}\hat{\beta}_0\\\hat{\beta}_1\end{bmatrix} + \hat{\boldsymbol{u}}$$



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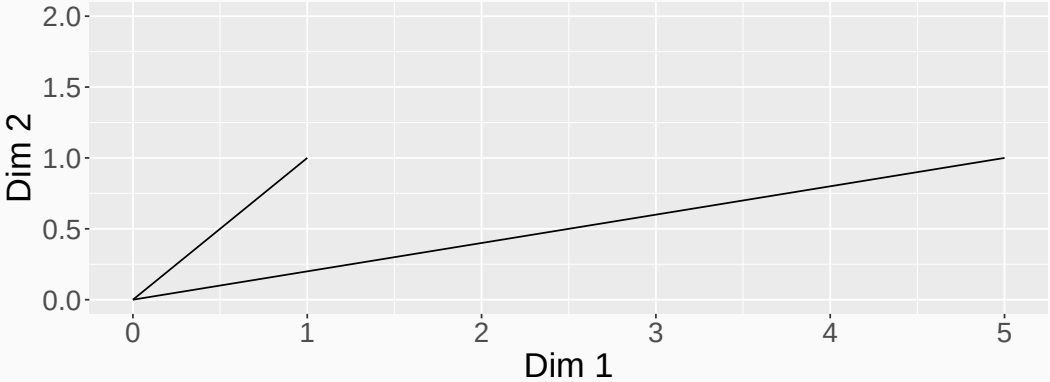

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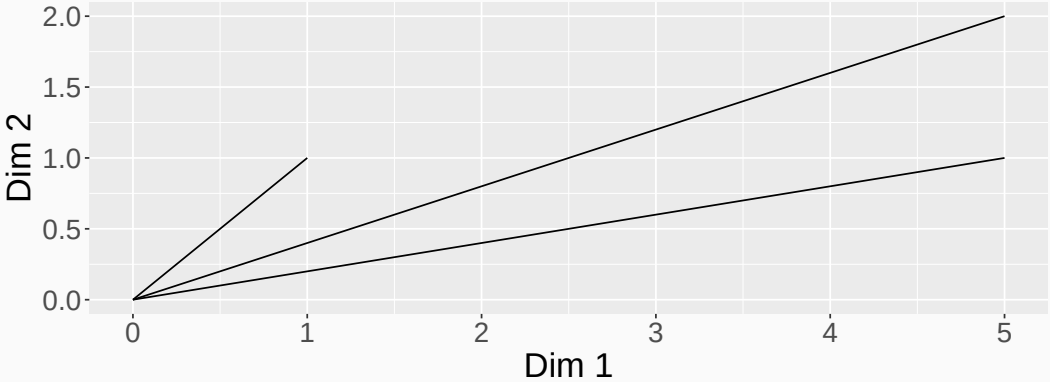

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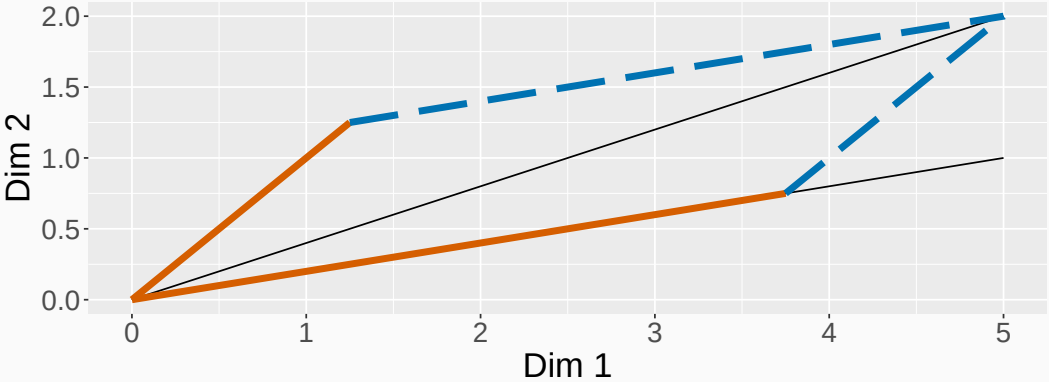

OLS geometry - 3 observations

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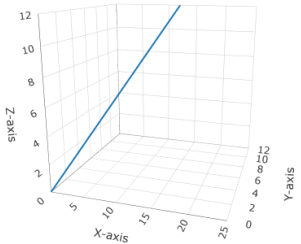

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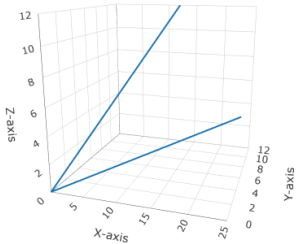

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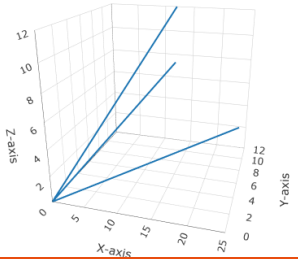

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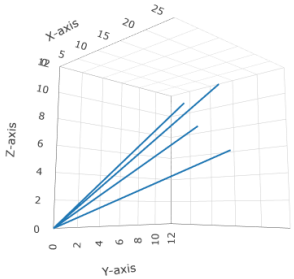

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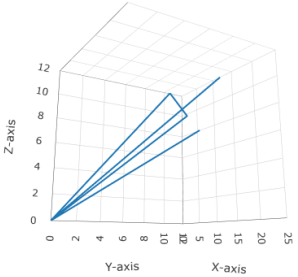

Forecast reconciliation as regression

We can write forecast reconciliation as

$$\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\beta_{t+h} + \boldsymbol{u}_{t+h}$$

where $\beta_{t+h} = \mathrm{E}(\boldsymbol{b}_{T+h}|\mathcal{I}_t)$.

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$$\hat{\boldsymbol{y}}_{t+h} = \boldsymbol{S}\beta_{t+h} + \boldsymbol{u}_{t+h}$$

where $\beta_{t+h} = \mathrm{E}(\boldsymbol{b}_{T+h}|\mathcal{I}_t)$. The OLS solution of this regression is

$$\hat{\beta}_{t+h} = (\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'\hat{\boldsymbol{y}}_{t+h},$$

which gives

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{S})^{-1}\boldsymbol{S}'$$

corresponding to the OLS estimator for the forecast error variance from before.

Forecast reconciliation

Coherent subspace

THe linear subspace $\mathfrak{s}$ for which the hierarchical structure hold for all $\boldsymbol{y} \in \mathfrak{s}$.

Forecast reconciliation

Coherent subspace

THe linear subspace $\mathfrak{s}$ for which the hierarchical structure hold for all $\boldsymbol{y} \in \mathfrak{s}$.

Coherent point forecasts

A vector of point forecast $\check{\boldsymbol{y}}_{t+h}$ of all series in the hierarchy is *coherent* if $\check{\boldsymbol{y}}_{t+h} \in \mathfrak{s}$.

Forecast reconciliation

Coherent subspace

THe linear subspace $\mathfrak{s}$ for which the hierarchical structure hold for all $\boldsymbol{y} \in \mathfrak{s}$.

Coherent point forecasts

A vector of point forecast $\check{\boldsymbol{y}}_{t+h}$ of all series in the hierarchy is *coherent* if $\check{\boldsymbol{y}}_{t+h} \in \mathfrak{s}$.

OLS forecast reconciliation

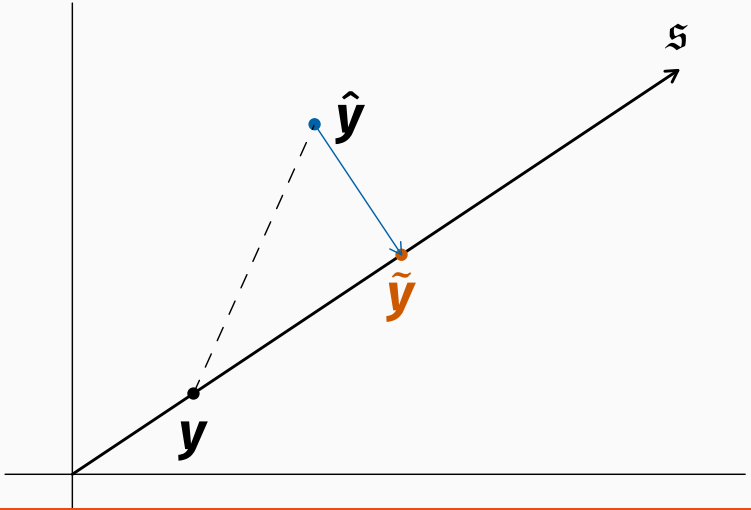
The OLS solution is an orthogonal projection of the base forecasts onto the coherent subspace $\mathfrak{s}$.

Distance-Reducing Property

If $\tilde{\boldsymbol{y}}_{t+h} = \boldsymbol{SG}\hat{\boldsymbol{y}}_{t+h}$, where $\boldsymbol{G}$ is such that $\boldsymbol{SG}$ is an orthogonal projection onto $\mathfrak{s}$, then

$$||\boldsymbol{y}_{t+h} - \tilde{\boldsymbol{y}}_{t+h}|| \leq ||\boldsymbol{y}_{t+h} - \hat{\boldsymbol{y}}_{t+h}||.$$

Orthogonal projection



Oblique projection - MinT

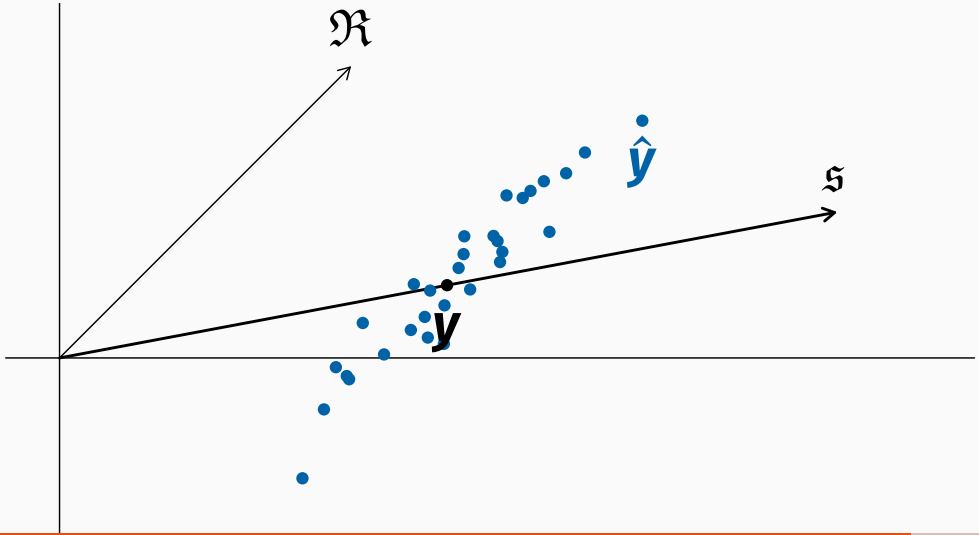
The MinT method is not an orthogonal projection. It is an oblique one.

MinT

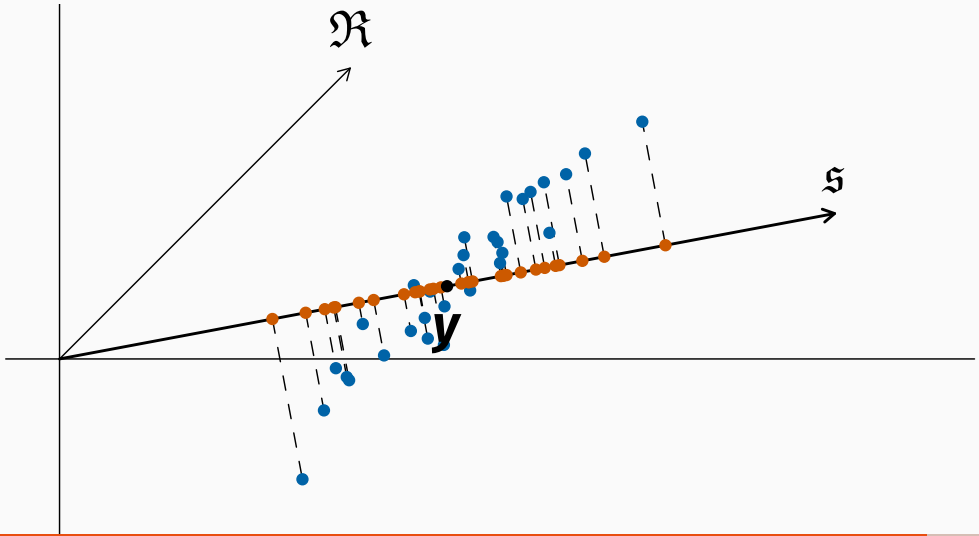
$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}$$

While orthogonal projection always improve on base forecasts, MinT minimises the loss function in **expectation** - minimum forecast error variances.

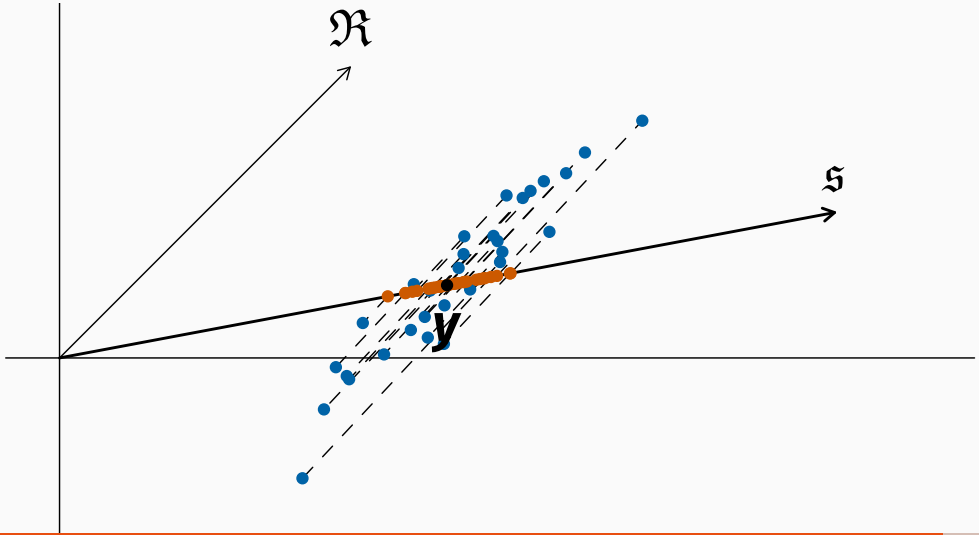
Base forecasts



Orthogonal projection



Oblique projection - MinT



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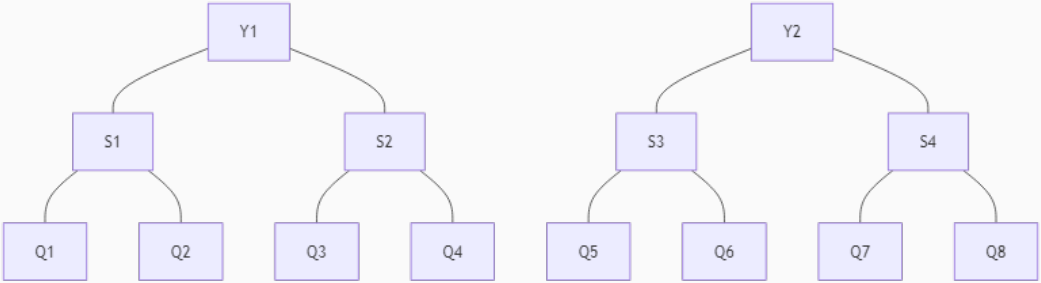
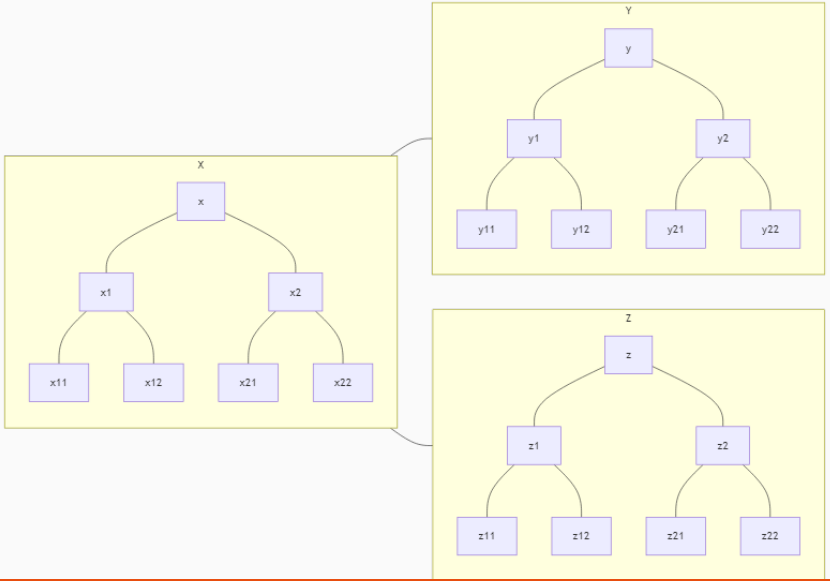
Temporal forecast reconciliation



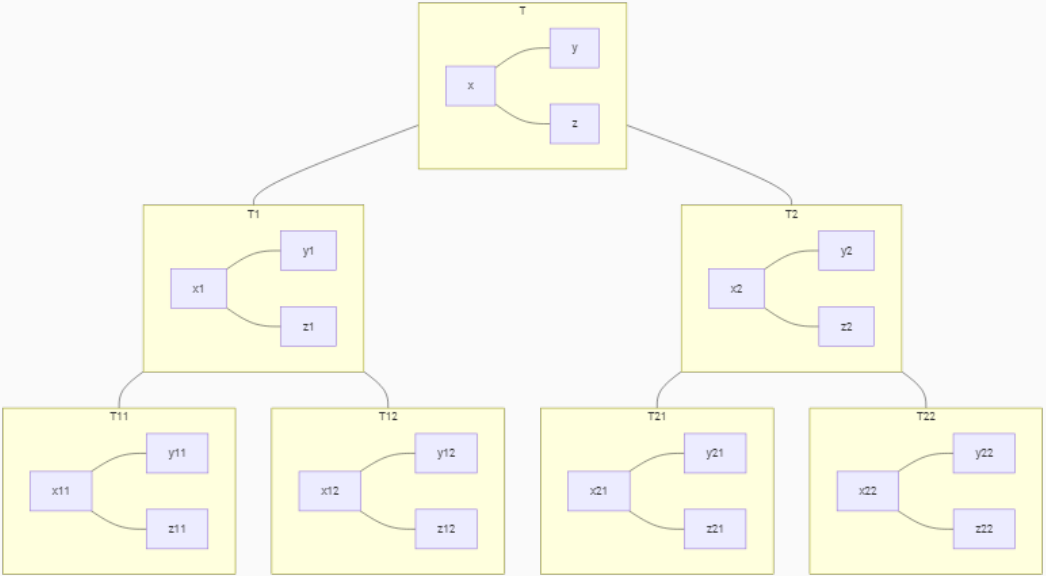
Temporal forecast reconciliation

- Single series observed at different frequencies
- Structural representation and zero constrained representation both still hold
- Very underused
  - Can be applied in most of the case as an extra step to improve accuracy
- See Athanasopoulos et al. (2017).

Cross-temporal forecast reconciliation



Cross-temporal forecast reconciliation



Cross-temporal forecast reconciliation

- Combining the two different hierarchies.
- More difficult to solve.
- Different algorithms and heuristics.
- See Di Fonzo and Girolimetto (2023)

Probabilistic forecast reconciliation

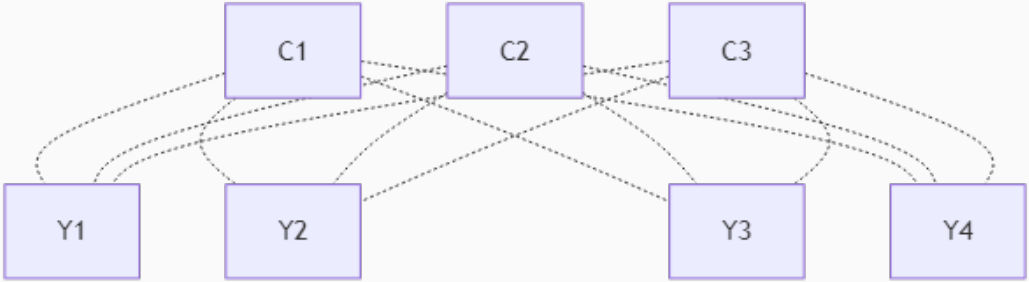
Forecast distribution

Apart from forecasting a single point, we can have a full forecast distribution.

Probabilistic forecast reconciliation

- Any possible forecasts that do not correctly add up according to the hierarchical structure are assigned zero probability.
- See Panagiotelis et al. (2023).

Forecast Linear Augmented Projection (FLAP)



FLAP

- Artificial upper level series
- Suitable for series even without a natural hierarchy
- Theoretical guarantee to reduce forecast error variance
  - The more artificial components the better
  - Optimal solution given a set of components
- Good empirical performance
- See Yang et al. (2024)

References

Ando, Sakai, and Futoshi Narita. 2024. "An Alternative Proof of Minimum Trace Reconciliation." *Forecasting* 6 (2): 456–61. https://doi.org/10.3390/forecast6020025.

Athanasopoulos, George, Rob J Hyndman, Nikolaos Kourentzes, and Fotios Petropoulos. 2017. "Forecasting with temporal hierarchies." *Eur. J. Oper. Res.* 262 (1): 60–74. https://doi.org/10.1016/j.ejor.2017.02.046.

Di Fonzo, Tommaso, and Daniele Girolimetto. 2023. "Cross-temporal forecast reconciliation: Optimal combination method and heuristic alternatives." *Int. J. Forecast.* 39 (1): 39–57. https://doi.org/10.1016/j.ijforecast.2021.08.004.

Opgen-Rhein, Rainer, and Korbinian Strimmer. 2007. "Accurate ranking of differentially expressed genes by a distribution-free shrinkage approach." *Stat. Appl. Genet. Mol. Biol.* 6 (1). https://doi.org/10.2202/1544-6115.1252.

Panagiotelis, Anastasios, George Athanasopoulos, Puwasala Gamakumara, and Rob J Hyndman. 2021. "Forecast reconciliation: A geometric view with new insights on bias correction." *Int. J. Forecast.* 37 (1): 343–59.